Monte Carlo Simulation Algorithm for Clustered N-of-1 Trial Designs

Mathematical Framework and Implementation Details

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Contents

1 Introduction

This white paper describes the complete algorithm for the clustered N-of-1 trial simulation framework comparing Hybrid and Open-Label with Blinded Discontinuation (OL+BDC) designs. The simulation evaluates statistical power to detect treatment × biomarker interactions under varying conditions of biomarker moderation (effect size modification) and carryover effects.

1.1 Motivation and Design Rationale

Traditional randomized controlled trials (RCTs) often fail to detect heterogeneous treatment effects when population-level biomarkers moderate the intervention response. N-of-1 designs offer an alternative by intensively measuring individual participants across multiple treatment periods. The clustered measurement schedule concentrates observations around critical transition points (e.g., treatment initiation, discontinuation) to maximize information density while maintaining numerical stability of the covariance matrix.

1.2 Simulation Scope

Designs evaluated: - Hybrid design: 4-path randomization with two treatment periods and measurement clusters at transitions - OL+BDC design: Open-label treatment followed by blinded discontinuation with measurement clusters

Parameter grid: - Biomarker moderation: 0 (null), 0.25, 0.35, 0.45, 0.55, 0.65 - Carryover retention: 0, 0.5, 1.0 (proportion of effect retained after discontinuation) - Sample size: 70 participants per iteration - Iterations: 1000 per condition

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2 Two-Stage Conditional Sampling Framework

2.1 Mathematical Justification for Simple Sigma Inversion

The fundamental challenge in multivariate data generation is constructing participant-level data that simultaneously respects: 1. Fixed correlation structure of latent response components 2. Relationship between biomarker level and baseline response 3. Conditional dependence of response trajectories on individual characteristics

We employ a two-stage algorithm that leverages properties of multivariate normal distributions to achieve efficient data generation with guaranteed consistency.

2.1.1 Block-Partitioned Covariance Structure

The complete covariance matrix is partitioned as:

$$\Sigma = \begin{pmatrix} \Sigma_{11} & \Sigma_{12} \\ \Sigma_{21} & \Sigma_{22} \end{pmatrix}$$

where:

- Σ_{11} is the $(3 \times n_{tp}) \times (3 \times n_{tp})$ covariance of latent response components: BR (biological response), ER (expectancy response), and TR (time response) across all timepoints
- Σ_{22} is the 2×2 covariance of biomarker and baseline response
- Σ_{12} is the $(3 \times n_{tp}) \times 2$ cross-covariance between responses and baseline characteristics
- $\Sigma_{21} = \Sigma_{12}^T$ by symmetry

Dimensions: $n_{tp} = 8$ measurement timepoints in clustered designs, yielding Σ of dimension $(3 \times 8 + 2) \times (3 \times 8 + 2) = 26 \times 26$.

2.1.2 Conditional Normal Distribution

For a partitioned multivariate normal random vector $(X_1, X_2) \sim N(\mu, \Sigma)$, the conditional distribution of X_1 given X_2 is:

$$X_1 | X_2 = x_2 \sim N(\mu_1 + \Sigma_{12}\Sigma_{22}^{-1}(x_2 - \mu_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21})$$

The key insight is that the **conditional covariance is independent of the observed value** x_2. This matrix, denoted Σ_{cond}, is computed once and reused across all participants:

$$\Sigma_{\text{cond}} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}^T$$

This is the **Schur complement** of Σ_{22} in Σ, computed via simple matrix operations: a single inversion of the 2×2 matrix Σ_{22}, which is computationally inexpensive and numerically stable.

2.2 Two-Stage Generation Algorithm

2.2.1 Stage 1: Generate Biomarker and Baseline

$$\begin{pmatrix} \text{Biomarker} \\ \text{Baseline} \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_{\text{BM}} \\ \mu_{\text{BL}} \end{pmatrix}, \Sigma_{22} \right)$$

with parameters: - $\mu_{\text{BM}} = 5.0$ (mean biomarker) - $\mu_{\text{BL}} = 10.0$ (mean baseline response) - $\Sigma_{22} = \begin{pmatrix} \sigma^2_{\text{BM}} & \rho_{\text{BM-BL}}\sigma_{\text{BM}}\sigma_{\text{BL}} \\ \rho_{\text{BM-BL}}\sigma_{\text{BM}}\sigma_{\text{BL}} & \sigma^2_{\text{BL}} \end{pmatrix}$
where $\sigma_{\text{BM}} = 2.0$, $\sigma_{\text{BL}} = 2.0$, $\rho_{\text{BM-BL}} = 0.25$.

2.2.2 Stage 2: Generate Response Components Conditional on Stage 1

Given observed biomarker b and baseline ℓ, generate response components:

$$\mathbf{r}|(b, \ell) \sim N(\mu_{\text{cond}}, \Sigma_{\text{cond}})$$

where the conditional mean is:

$$\mu_{\text{cond}} = \Sigma_{12}\Sigma_{22}^{-1} \begin{pmatrix} b - \mu_{\text{BM}} \\ \ell - \mu_{\text{BL}} \end{pmatrix}$$

and the conditional covariance is:

$$\Sigma_{\text{cond}} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{12}^T$$

2.2.3 Justification for Simple Inversion

The inversion of Σ_{22} is justified because:

1. **Dimension reduction**: Σ_{22} is only 2×2, making inversion trivial and highly stable. The closed form is:

$$\Sigma_{22}^{-1} = \frac{1}{\det(\Sigma_{22})} \begin{pmatrix} \sigma^2_{\text{BL}} & -\rho_{\text{BM-BL}}\sigma_{\text{BM}}\sigma_{\text{BL}} \\ -\rho_{\text{BM-BL}}\sigma_{\text{BM}}\sigma_{\text{BL}} & \sigma^2_{\text{BM}} \end{pmatrix}$$

   where $\det(\Sigma_{22}) = \sigma^2_{\text{BM}}\sigma^2_{\text{BL}}(1 - \rho^2_{\text{BM-BL}})$.

2. **Numerical stability**: For $\rho_{\text{BM-BL}} = 0.25$, $\det(\Sigma_{22}) \approx 3.75 > 0$, ensuring well-conditioned inversion (condition number $\kappa \ll 100$).

3. **Computational efficiency**: A single 2×2 inversion per participant is negligible compared to the cost of multivariate normal generation.

4. **Mathematical correctness**: The Schur complement formula is exact under joint normality, with no approximation error. The conditional covariance is positive definite by construction when Σ is positive definite.

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3 Sigma Matrix Construction

3.1 AR(1) Time-Based Autocovariance

Response components exhibit temporal autocorrelation modeled as continuous-time AR(1):

$$\text{Cov}(Y_i(t), Y_i(t')) = \sigma^2 \rho^{|t-t'|}$$

where $\rho \in [0, 1)$ is the correlation coefficient and $|t - t'|$ is the absolute time difference in weeks.

For each response component, we construct an $n_{tp} \times n_{tp}$ autocovariance matrix:

$$\Sigma_{\text{component}} = \begin{pmatrix} \sigma^2 & \sigma^2 \rho^{|t_1-t_2|} & \cdots & \sigma^2 \rho^{|t_1-t_{n_{tp}}|} \\ \sigma^2 \rho^{|t_2-t_1|} & \sigma^2 & \cdots & \sigma^2 \rho^{|t_2-t_{n_{tp}}|} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma^2 \rho^{|t_{n_{tp}}-t_1|} & \cdots & \cdots & \sigma^2 \end{pmatrix}$$

Parameters:

Component	ρ	σ	Purpose
BR (biological)	0.75	1.8	Drug effect accumulation
ER (expectancy)	0.75	1.8	Placebo/expectancy effect
TR (time-variant)	0.75	1.8	Natural improvement over time

3.2 Cross-Component Covariance

Between different response components, covariance depends on temporal proximity:

$$\text{Cov}(Y_i^{(c_1)}(t), Y_i^{(c_2)}(t')) = \begin{cases} \gamma_{1t} & \text{if } t = t' \\ \gamma_{ct} \rho_{\text{decay}}^{|t-t'|} & \text{if } t \neq t' \end{cases}$$

where: - $\gamma_{1t} = 0.12 \times \sigma^2$ (same-timepoint cross-correlation) - $\gamma_{ct} = 0.05 \times \sigma^2$ (different-timepoint cross-correlation) - $\rho_{\text{decay}} = 0.9$ (time decay factor)

This structure ensures that response components share information most strongly at the same measurement point, with correlation decaying exponentially with time separation.

3.3 Cross-Covariance with Baseline Characteristics

The Σ_{12} matrix (responses vs. biomarker/baseline) encodes how biomarker and baseline response predict individual response trajectories:

$$\Sigma_{12} = \begin{pmatrix} \mathbf{c}_{\text{BR,BM}} & \mathbf{c}_{\text{BR,BL}} \\ \mathbf{c}_{\text{ER,BM}} & \mathbf{c}_{\text{ER,BL}} \\ \mathbf{c}_{\text{TR,BM}} & \mathbf{c}_{\text{TR,BL}} \end{pmatrix}$$

where each row vector contains covariances between one response component and the two baseline characteristics:

- $\mathbf{c}_{\text{BR,BM}} = c_{\text{BM}} \times \sigma_{\text{BR}} \times \sigma_{\text{BM}} \times \mathbf{1}_{n_{tp}}$ with $c_{\text{BM}} = 0.3$ (biomarker correlation, varies in parameter grid)
- $\mathbf{c}_{\text{ER,BM}} = 0.5 \times c_{\text{BM}} \times \sigma_{\text{ER}} \times \sigma_{\text{BM}} \times \mathbf{1}_{n_{tp}}$ (ER is half as sensitive to biomarker)
- $\mathbf{c}_{\text{TR,BM}} = 0.5 \times c_{\text{BM}} \times \sigma_{\text{TR}} \times \sigma_{\text{BM}} \times \mathbf{1}_{n_{tp}}$ (TR is half as sensitive to biomarker)
- $\mathbf{c}_{*,\text{BL}} = c_{\text{BL}} \times \sigma_{*} \times \sigma_{\text{BL}} \times \mathbf{1}_{n_{tp}}$ with $c_{\text{BL}} = 0.3$ (baseline correlation, fixed)

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4 Carryover Effect Modeling

4.1 Current Implementation: One-Week Carryover Decay

When a participant transitions from treatment (on-drug) to placebo/control (off-drug), the biological effect decays exponentially. The current model implements a simple one-week carryover window:

$$\text{BR}_{\text{mean},i,t} = \begin{cases} w_t^{\text{on}} \times \text{effect}_i(t) & \text{if treatment}_i(t) = 1 \\ \text{carryover} \times \text{BR}_{\text{accum},i,t-1} & \text{if treatment}_i(t) = 0 \text{ and treatment}_i(t-1) = 1 \\ 0 & \text{if treatment}_i(t) = 0 \text{ and treatment}_i(t-1) = 0 \end{cases}$$

where:

- w_t^{on} = cumulative weeks on treatment at week t
- $\text{effect}_i(t) = \text{BR}_{\text{rate}} \times (1 + \text{bm\_mod} \times z_i)$ is the biomarker-moderated treatment effect
- bm_mod is the biomarker moderation parameter (0 to 0.65)
- $z_i = (\text{biomarker}_i - \overline{\text{biomarker}})/\sigma_{\text{biomarker}}$ is standardized biomarker
- $\text{BR}_{\text{accum},i,t-1}$ is the accumulated BR effect just before discontinuation
- $\text{carryover} \in \{0, 0.5, 1.0\}$ is the proportion of previous effect retained

4.1.1 Biological Interpretation

The carryover parameter represents the **half-life of the drug effect** after discontinuation. For example: - carryover = 0: No residual effect (clean wash-out at discontinuation) - carryover = 0.5: 50% of the previous week's effect persists - carryover = 1.0: Full effect persists (no decay)

The effect applies **only to the immediate post-treatment week** in the current implementation.

4.2 Alternative 1: Exponential Decay with Half-Life Parameter

Instead of a fixed one-week window, model carryover as exponential decay:

$$\text{BR}_{\text{mean},i,t} = \begin{cases} w_t^{\text{on}} \times \text{effect}_i(t) & \text{if treatment}_i(t) = 1 \\ \text{BR}_{\text{accum},i,t_{\text{disc}}} \times 0.5^{\frac{t-t_{\text{disc}}}{t_{1/2}}} & \text{if } t > t_{\text{disc}} \end{cases}$$

where:

- t_{disc} is the discontinuation timepoint
- $t_{1/2}$ is the **half-life in weeks** (varies in parameter grid: 0, 1.0, 2.0, 3.0)

- The decay rate $0.5^{\frac{\Delta t}{t_{1/2}}}$ reflects proportion remaining after Δt weeks

4.2.1 Implementation Details

For clustered designs with 8 measurement points, carryover window spans multiple weeks. Example (Hybrid design, discontinuation at week 12):

Week	Treatment	Time off	Decay factor	BR contribution
8	ON	0	1.0	$w_8 \times \text{effect}_i$
9	OFF	1	$0.5^{1/t_{1/2}}$	accum $\times\ 0.5^{1/t_{1/2}}$
10	OFF	2	$0.5^{2/t_{1/2}}$	accum $\times\ 0.5^{2/t_{1/2}}$
11	OFF	3	$0.5^{3/t_{1/2}}$	accum $\times\ 0.5^{3/t_{1/2}}$
12	OFF	4	$0.5^{4/t_{1/2}}$	accum $\times\ 0.5^{4/t_{1/2}}$

4.2.2 Advantages

- **Biologically realistic**: Matches pharmacokinetic elimination kinetics
- **Flexible duration**: Single parameter ($t_{1/2}$) controls entire carryover curve
- **Continuous decay**: Captures effect persistence across all post-treatment measurements
- **Identifiable**: Each half-life value produces distinct power curves

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4.3 Alternative 2: Extended Two-Week Carryover Window

Instead of exponential decay, extend the carryover effect to persist across **two weeks** after discontinuation:

$$\text{BR}_{\text{mean},i,t} = \begin{cases} w_t^{\text{on}} \times \text{effect}_i(t) & \text{if treatment}_i(t) = 1 \\ \text{carryover} \times \text{BR}_{\text{accum},i,t-1} & \text{if } t = t_{\text{disc}} + 1 \text{ (week 1 post-disc)} \\ \text{carryover} \times \text{carryover}_2 \times \text{BR}_{\text{accum},i,t-2} & \text{if } t = t_{\text{disc}} + 2 \text{ (week 2 post-disc)} \\ 0 & \text{if } t > t_{\text{disc}} + 2 \end{cases}$$

where:

- $\text{carryover}_2 \in [0, 1]$ is an additional decay factor for the second week
- Typical choice: $\text{carryover}_2 = 0.5 \times \text{carryover}$ (50% of first-week decay)

4.3.1 Implementation Details

Example with carryover $= 0.7$ and $\text{carryover}_2 = 0.35$:

Week	Treatment	Week 1 factor	Week 2 factor	BR contribution
8	ON	—	—	$w_8 \times \text{effect}_i$
9	OFF	0.7	—	accum $\times$ 0.7
10	OFF	—	0.35	accum $\times$ 0.35
11	OFF	0	0	0

Week	Treatment	Week 1 factor	Week 2 factor	BR contribution
12	OFF	0	0	0

4.3.2 Advantages

- **Simple parameterization**: Two intuitive parameters (carryover, carryover_2)
- **Bounded support**: Effect explicitly ends after two weeks (matches trial design)
- **Captures biphasic decay**: First-week effect typically larger than second-week
- **Computationally efficient**: No exponentials; simple conditional logic

4.3.3 Comparison of Approaches

Aspect	Current (1-Week)	Alternative 1 (Exp. Decay)	Alternative 2 (2-Week)
Carryover window	1 week	Multiple weeks	2 weeks
Decay mechanism	Step function	Exponential	Linear steps
Parameters	carryover	$t_{1/2}$	carryover, carryover
Biological plausibility	Low	High	Medium
Computational cost	Minimal	Exponentiations	Minimal
Power sensitivity	Moderate	High	Moderate-High
Matched trials	Clustered PD designs	Pharmacokinetic studies	Discontinuation RCTs

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5 Biomarker Interaction Generation and Power Calculation

5.1 Two-Level Interaction Mechanism

The treatment × biomarker interaction is generated through two complementary mechanisms that operate at different levels of the data-generating process:

5.1.1 Level 1: Covariance-Level Interaction (Structural)

The cross-covariance between biomarker and biological response (Σ_{12}) induces correlation between individual biomarker levels and baseline response variability. Individuals with higher biomarkers systematically have higher baseline responses and, through the conditional normal structure, more variable response trajectories:

$$\text{Cov}(\text{BR}_i(t), \text{Biomarker}_i) = c_{\text{BM}} \times \sigma_{\text{BR}} \times \sigma_{\text{BM}}$$

This generates **individual-level heterogeneity** but does **not** by itself create an interaction with treatment in a regression model.

5.1.2 Level 2: Mean-Level Moderation (Functional)

The treatment effect size depends on biomarker level:

$$\text{effect}_i(t) = \text{BR}_{\text{rate}} \times (1 + \text{bm\_mod} \times z_i)$$

where $z_i = \frac{\text{biomarker}_i - \overline{\text{biomarker}}}{\sigma_{\text{biomarker}}}$.

This **moderation function** makes the treatment effect proportional to biomarker level. As weeks accumulate on treatment:

$$\text{BR}_{\text{mean},i,t} = w_t^{\text{on}} \times \text{BR}_{\text{rate}} \times (1 + \text{bm\_mod} \times z_i)$$

The interaction emerges in the mixed-effects model because: 1. Treatment accumulates (increases w_t^{on}) 2. But the **slope of effect accumulation** varies with biomarker

5.2 Mathematical Formulation of the Interaction

The full model for biological response mean at week t is:

$$E[\text{BR}_i(t) | \text{treatment}_i(t), z_i] = (w_t^{\text{on}} \times \text{BR}_{\text{rate}}) \times (1 + \text{bm\_mod} \times z_i)$$

Expanding:

$$= w_t^{\text{on}} \times \text{BR}_{\text{rate}} + w_t^{\text{on}} \times \text{BR}_{\text{rate}} \times \text{bm\_mod} \times z_i$$

The second term is the **treatment × biomarker interaction at the mean level**:

$$\text{Interaction effect} = w_t^{\text{on}} \times \text{BR}_{\text{rate}} \times \text{bm\_mod} \times z_i$$

In the linear mixed-effects model fitted to the data:

$$\text{response}_{i,t} = \beta_0 + \beta_{\text{treatment}} \times \text{treatment}_{i,t} + \beta_{\text{bm}} \times \text{bm\_centered}_i + \beta_{\text{int}} \times \text{treatment}_{i,t} \times \text{bm\_centered}_i + \beta_{\text{week}} \times \text{week}_t + u$$

the interaction coefficient β_{int} captures the **average slope change in treatment effect per unit biomarker difference**.

5.3 Power as a Function of Interaction Size

Power to detect the interaction (= 0.05, two-sided test) depends on:

1. **Effect size magnitude**: bm_mod $\in \{0, 0.25, 0.35, 0.45, 0.55, 0.65\}$
    - Larger values make the interaction more pronounced
    - bm_mod = 0 represents the **null case** (type I error condition)
    - bm_mod = 0.65 represents the **largest effect**
2. **Treatment accumulation window**: Clustered designs with 8 timepoints enable w_t^{on} to reach 8 weeks on treatment, amplifying the interaction effect through multiplication
3. **Carryover attenuation**: When carryover > 0, discontinuation reduces the final treatment accumulation and thus reduces interaction detectability
4. **Sample size and iterations**: 70 participants, 1000 iterations per condition provides stable power estimates with standard error < 0.015

5.4 Null Hypothesis and Type I Error Control

When bm_mod = 0 (null condition), the treatment effect is **independent of biomarker**:

$$\text{effect}_i(t) = \text{BR}_{\text{rate}} \times (1 + 0 \times z_i) = \text{BR}_{\text{rate}}$$

The interaction coefficient β_{int} follows a **t-distribution under the null hypothesis** with degrees of freedom   nparticipants $\times$ ntimepoints $-$ nfixed effects $= 70 \times 8 - 4 = 556$.

Type I error rate is evaluated by computing power under the null:

$$\text{Type I error} = P(\text{p-value} < 0.05 | \text{bm\_mod} = 0)$$

This should be approximately 0.05 for a valid test.

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6 Monte Carlo Simulation Algorithm

6.1 Overall Flow

```
FOR each condition in parameter_grid:
  FOR iter = 1 to n_iterations:

    1. GENERATE TRIAL DESIGN
       - Assign 70 participants to 4-path randomization
       - Define treatment indicators across 8 clustered measurement weeks

    2. BUILD COVARIANCE STRUCTURE
       - Construct Sigma (26 × 26 matrix)
       - Compute Schur complement: Sigma_cond = Sigma_11 - Sigma_12 @ Sigma_22^-1 @ Sigma_12^T
```

```
   3. GENERATE PARTICIPANT DATA (two-stage)
      FOR each participant i:
        - Stage 1: Generate (biomarker_i, baseline_i) ~ MVN(mu_22, Sigma_22)
        - Stage 2: Generate (BR_i, ER_i, TR_i) | (biomarker_i, baseline_i) ~ MVN(mu_cond, Sigm

   4. COMPUTE RESPONSE TRAJECTORY
      FOR each participant i and week t:
        - Compute treatment × biomarker moderated effect
        - Add accumulated BR, ER, TR components
        - Incorporate carryover decay if applicable
        - response_i,t = baseline_i + BR_mean + BR_random + ER_mean + ER_random + TR_mean + TR

   5. FIT MIXED-EFFECTS MODEL
      - Fit lme() with random intercepts and AR(1) correlation structure
      - Extract interaction term coefficient and p-value

   6. RECORD RESULT
      - significant = (p-value < 0.05)
      - Store effect size, SE, t-value, p-value

 AGGREGATE across 1000 iterations:
 - Power = proportion of iterations with p < 0.05
 - Mean effect size and SD
```

6.2 Pre-Simulation Validation Phase

Before running the expensive Monte Carlo loop, validate that the covariance matrices are positive definite:

```
FOR each unique sigma structure (design, measurement_weeks, c.bm):
  1. Build sigma matrix using build_sigma_guaranteed_pd()
  2. Compute eigenvalues of full Sigma matrix
  3. Check: min(eigenvalues) > 1e-6 (positive definiteness)
  4. Compute condition number   = max(eigenvalues) / min(eigenvalues)
  5. Warn if   > 100 (ill-conditioned)

IF any sigma fails validation:
  STOP with diagnostic error
ELSE:
  Proceed to Monte Carlo simulation
```

6.3 Mixed-Effects Model Specification

The regression model fitted to each simulated dataset includes:

$$\text{response}_{i,t} = \beta_0+\beta_{\text{treatment}}\times\text{treatment}_{i,t}+\beta_{\text{bm\_centered}}\times\text{bm\_centered}_i+\beta_{\text{int}}\times\text{treatment}_{i,t}\times\text{bm\_centered}_i+\beta_{\text{week}}\times\text{w}$$

where:

- $b_i \sim N(0, \tau^2)$ is a random intercept for participant i
- $\epsilon_{i,t}$ follows an AR(1) correlation structure: $\text{Corr}(\epsilon_{i,t}, \epsilon_{i,t'}) = \phi^{|t-t'|}$
- $\text{bm\_centered}_i = \text{biomarker}_i - \overline{\text{biomarker}}$ is centered at the sample mean

Hypothesis test: $H_0 : \beta_{\text{int}} = 0$ vs. $H_1 : \beta_{\text{int}} \neq 0$

Test statistic: $t = \frac{\hat{\beta}_{\text{int}}}{\widehat{\text{SE}}(\hat{\beta}_{\text{int}})} \sim t_{556}$

Power calculation: Power = proportion of 1000 iterations where $|t| > t_{0.975,556}$ or equivalently p-value < 0.05.

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7 Results Aggregation and Interpretation

7.1 Summary Statistics

For each condition (combination of design, biomarker_moderation, carryover), the 1000 iterations produce a distribution of results. Summary statistics computed:

Power

$$\text{Power} = \frac{\sum_{j=1}^{1000} I(p_j < 0.05)}{1000}$$

where $I(\cdot)$ is the indicator function.

Mean Effect Size

$$\overline{\beta}_{\text{int}} = \frac{1}{1000} \sum_{j=1}^{1000} \hat{\beta}_{\text{int},j}$$

Effect Size Variability

$$\hat{\sigma}_{\beta_{\text{int}}} = \sqrt{\frac{1}{999} \sum_{j=1}^{1000} (\hat{\beta}_{\text{int},j} - \overline{\beta}_{\text{int}})^2}$$

7.2 Power Heatmap Visualization

Results are visualized as a 2D heatmap with: - **Rows**: Biomarker moderation levels (0 to 0.65) - **Columns**: Carryover retention levels (0, 0.5, 1.0) - **Cell color**: Power (0% red to 100% green) - **Facets**: Separate panels for Hybrid and OL+BDC designs

This visualization reveals: 1. **Main effect of biomarker moderation**: Steeper slopes → higher power 2. **Carryover penalty**: Higher carryover (retention) → higher power 3. **Design comparison**: Which design achieves higher power for same conditions 4. **Type I error**: Cell at bm_mod = 0 should show 5% power

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8 Computational Considerations

8.1 Complexity Analysis

Sigma matrix construction: $O(n_{tp}^2) = O(64)$ per participant per iteration - AR(1) autocovariance matrices: 3 matrices of dimension 8×8 - Cross-covariance filling: 8×8 nested loops with simple arithmetic - Schur complement: single inversion of 2×2 matrix

Data generation: $O(n_{tp})$ per participant per iteration - Stage 1: 2D multivariate normal generation - Stage 2: 24D multivariate normal generation (3×8) - Response trajectory computation: 8 weeks per participant

Model fitting: $O(n_{tp}^3)$ per iteration (general mixed-effects) - Matrix operations in REML/ML estimation dominate - 560 observations (70 participants × 8 timepoints) per iteration

Total per condition: ~5–15 seconds for 1000 iterations on modern hardware (M1 Mac)

8.2 Pre-Simulation Validation Speedup

Testing 3 sigma structures before launching full simulation prevents scenarios where: - All 1000 iterations fail to fit due to non-PD matrix (wasted computation) - Eigenvalue problems only appear during parameter sweeps (caught early)

Validation phase: <1 second per condition. Saved computation if failure: 15 min.

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9 References

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10 Appendices

10.1 Appendix A: Clustered Measurement Schedules

10.1.1 Hybrid Design (8 weeks)

Weeks: 4, 8, 9, 10, 11, 12, 16, 20 - Cluster 1: Weeks 8–12 (treatment transition) - Sparse coverage: Baseline (week 4), follow-up (weeks 16, 20)

10.1.2 OL+BDC Design (8 weeks)

Weeks: 4, 8, 12, 16, 17, 18, 19, 20 - Cluster 1: Weeks 17–20 (discontinuation) - Sparse coverage: Baseline (week 4), open-label (weeks 8, 12, 16)

Both designs maximize information density at critical transition points while maintaining positive definiteness of the covariance matrix.

10.2 Appendix B: Software Implementation

All simulations implemented in R 4.4+ using: - `MASS::mvrnorm()` for multivariate normal generation - `nlme::lme()` for mixed-effects models - `tidyverse` for data wrangling and visualization - `ggplot2` for publication-quality heatmaps

Pre-simulation validation code available in `pm_functions.R`: `validate_parameter_grid()` and `build_sigma_guaranteed_pd()`.